



Improving Bird Image Classification By Combining Visual And Language-Based Learning Approaches

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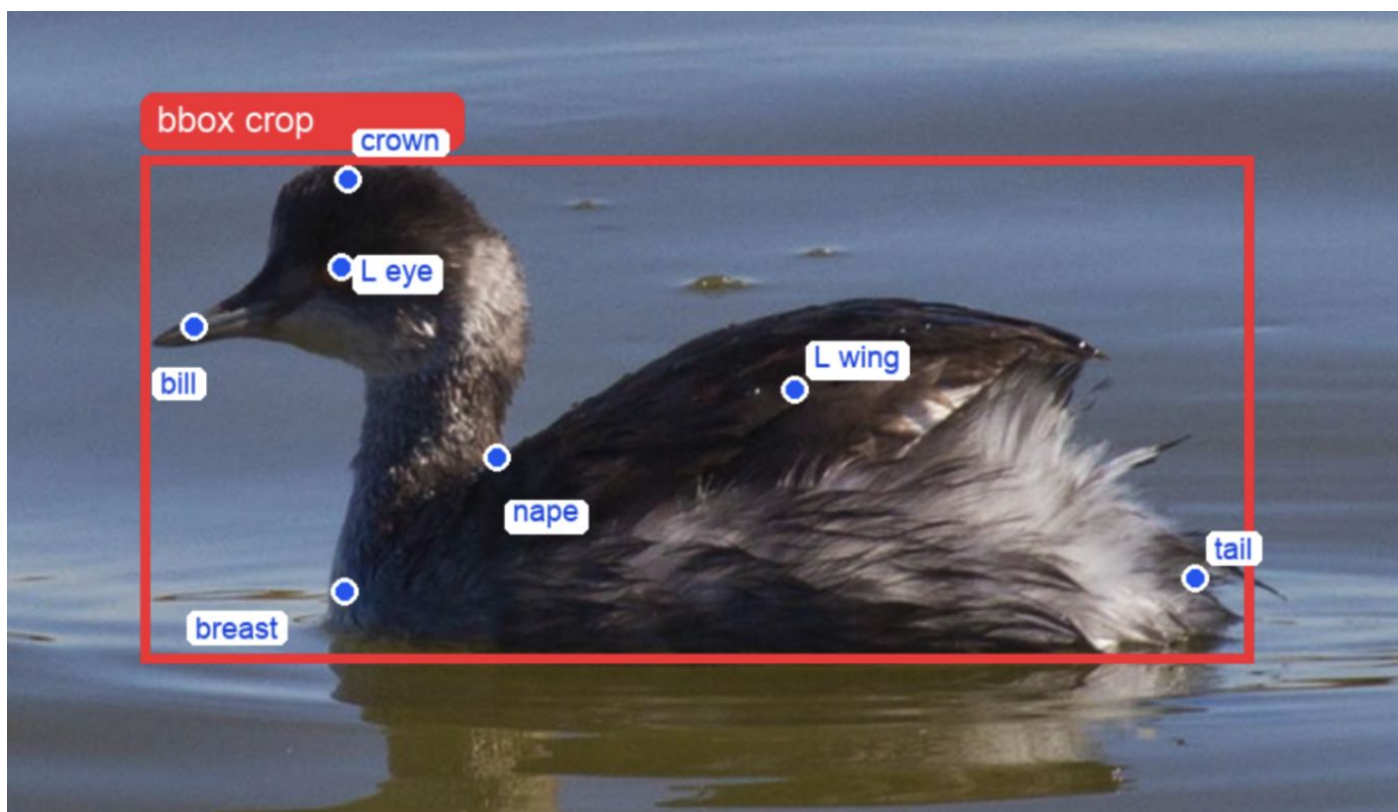
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Project Overview

- Motivation:** Fine-grained bird species classification from images is essential for ecological conservation, wildlife monitoring, and biological research, but is a difficult task
- Challenges:**
 - High inter-class similarity (different species are visually similar)
 - High intra-class variation (individuals of same species can look different)
- Prior work directions:**
 - Visual-only** methods with *enhancements* (attention, feature selection/enhancement)
 - Multimodal** methods (vision, language, metadata, habitat data, etc.)
- Our Idea:** Combine *vision* and *language* approaches
- Problem statement:**
 - Input:** 2D RGB image of a bird (+ optional text: class name/visual description)
 - Output:** predicted species class $c \in \{1, \dots, C\}$

Dataset & Metrics

- NABirds:** 48,562 images of 555 North American bird species
 - 20,339 Train/3,590 Validation/24,633 Test
 - Contains bounding boxes and part annotations
- Example image:



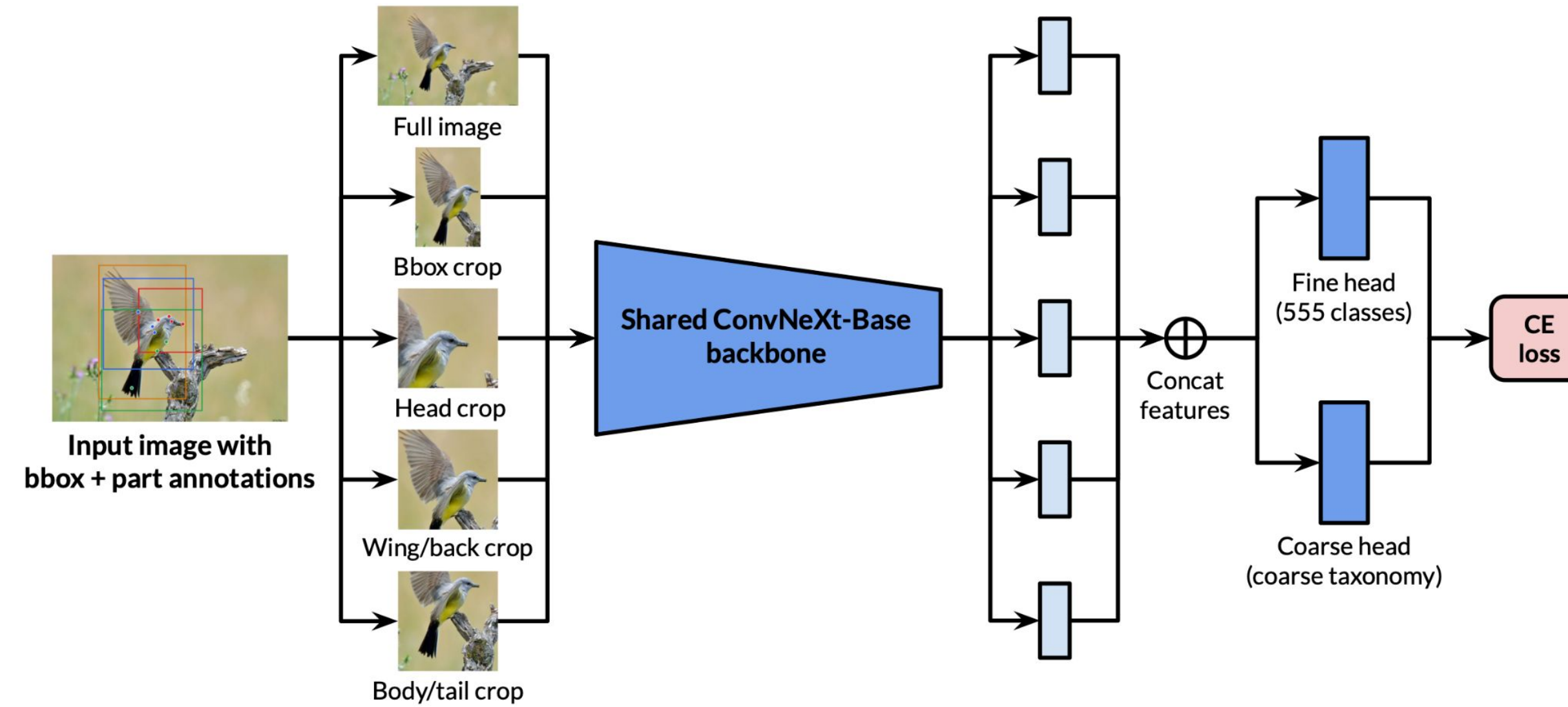
Class label: Eared Grebe (Nonbreeding/juvenile)

- Evaluation metrics:
 - Top-k accuracy, $k \in \{1, 5\}$
 - Macro-F1 score:

$$\text{Macro-F1} = \frac{1}{C} \sum_{k=1}^C \frac{2P_k R_k}{P_k + R_k}$$

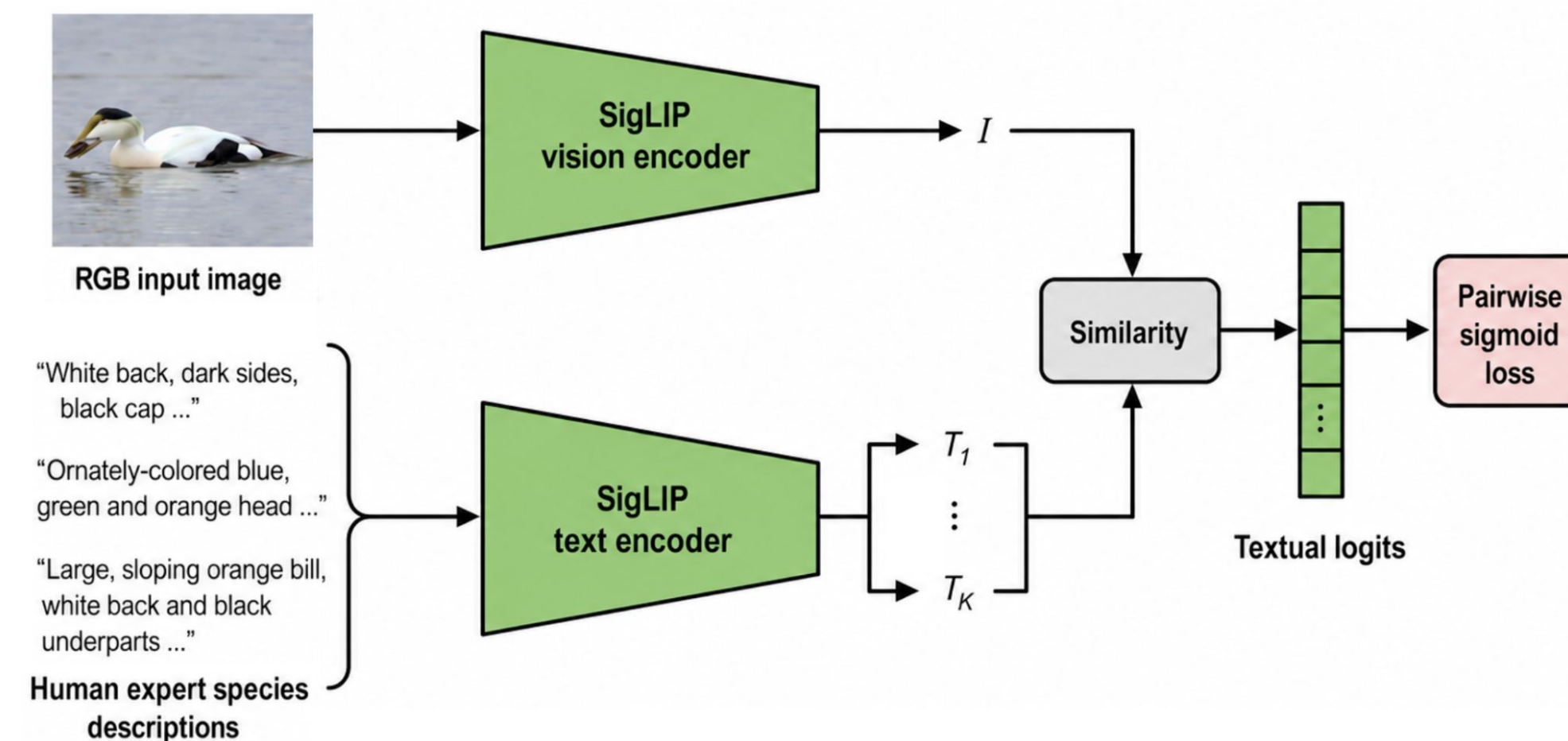
Methods & Experiments

Visual-only approach:



Training: ConvNeXt backbone + classification heads are fine-tuned end-to-end.

Vision-language approach:



SigLIP training: Vision side trained, text side frozen and optimized with pairwise sigmoid loss.

Fusion approach:

$$s_i = \lambda \log \text{softmax}(\ell_i^V) + (1 - \lambda) \log \text{softmax}(\ell_i^T)$$

Discussions & Future Research

Observations:

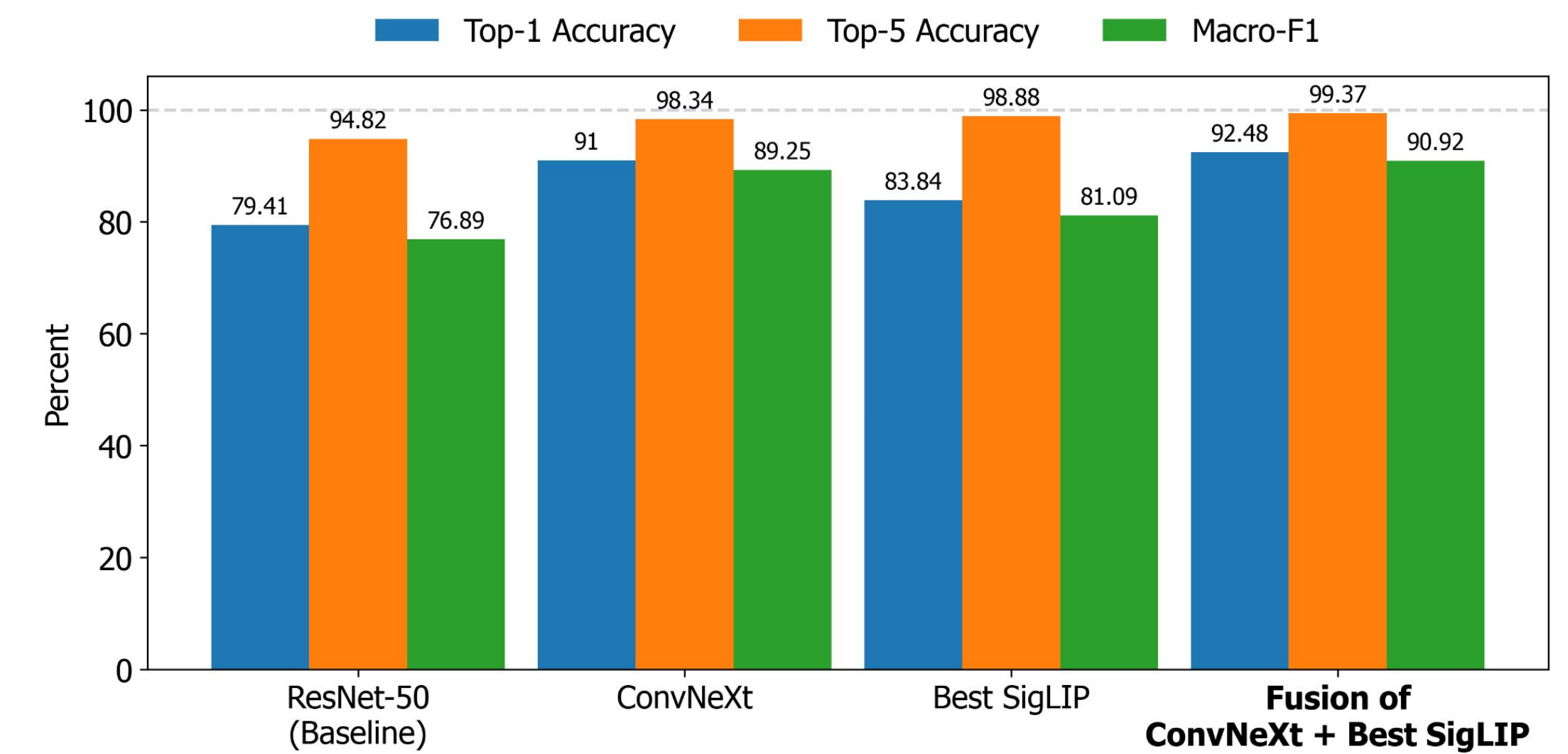
- Combining visual-only + text-based methods with lightweight fusion improves accuracy compared to individual methods → Fusion top-1 accuracy is **close to state-of-the-art!**
- Class label alone is best text input, additional visual descriptions hurt performance
- Misclassifications are still reasonable (visually similar classes)

Future Research:

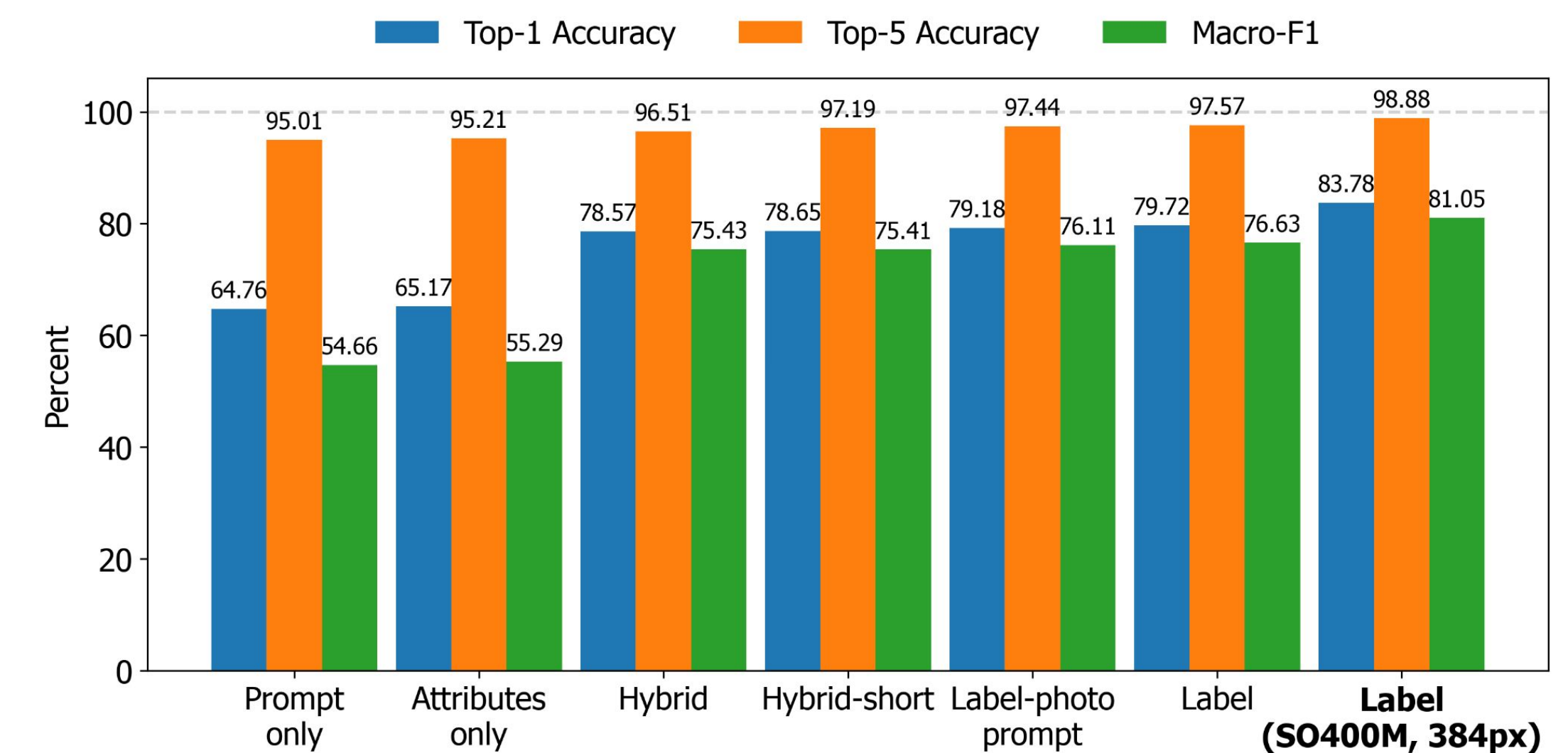
- Cleaning/refining descriptions, using other types of VLMs, trying new fusion approaches

Results

Overall results:



SigLIP ablation study:



Misclassification Examples	Ground Truth Class	Prediction Class
SigLIP Misclassifications	Hutton's Vireo	Ruby-crowned Kinglet
ConvNeXt Misclassifications	Hairy Woodpecker	Downy Woodpecker